

For example:

```
tar -C /usr/local -xzf go1.4.linux-amd64.tar.gz
```

Add `/usr/local/go/bin` to the `PATH` environment variable, as shown below:

```
export PATH = $PATH:/usr/local/go/bin
```

For installing Go in Ubuntu, run the following command:

```
sudo apt-get install golang
```

After the Go installation is done, you can check the version by using the following command:

```
go version
```

Programming with Go

Some of Google's sample Go code reveals that the syntax is C-like. It incorporates functions, decision making and loop control statements, standard conditional expressions and many other features that you'd expect to find in a C-like language, but with a handful of nice twists. There are some Python-like features too, including array slices and a map type with constructor syntax that looks like Python's dictionary concept but eschews some typical object-oriented programming concepts such as inheritance.

A Go program can vary in length from three lines to millions of lines, and it should be written into one or more text files with the extension `.go`. For example, `hello.go`.

Vi, vim or any other text editor too can be used to write a Go program into a file. Ensure the Go compiler is in the path and that it is being run in the directory containing the source file `hello.go`.

Our first Go program: *first.go*

To begin, give the following commands:

```
package main
import "fmt"
func main()
{
    fmt.Println("Hello World!")
}
```

Compile it and run it using the `go run` command:

```
go run first.go
```

Here is the output:

```
Hello World!
```

Here is a brief explanation of what each line does in the program.

package main: This is the first line of the program and defines the name of the package in which this program lies. It is a mandatory statement, as Go programs run in packages that provide code reusability.

import fmt: This is a preprocessor command, which tells the Go compiler to include the files in the package `fmt`.

func main(): This is the main function where the program execution begins.

fmt.Println(): The `fmt` package has exported the `Println` method, which is used to write text to the standard output. P is written in upper case. In the Go language, if a name that starts with a capital letter is exported, it implies that the function or variable/constant is accessible to the importer of the respective package.

Tokens in Go

Tokens are the building blocks of any programming language and consist of keywords, identifiers, variables and constants. There are 25 keywords in Golang and some of the reserved words are: `break`, `chan`, `defer`, `fallthrough`, `go`, `map`, `package`, `var`, `type`, `interface`, etc. An identifier in Go starts with any letter from A to Z or a to z or an underscore (`_`) followed by zero or more letters, underscores, and digits 0 to 9.

Data types in Go

The following are the basic data types available in Go:

1. Boolean types that consist of the two predefined constants — (a) `true`, and (b) `false`.
2. Numeric types that are again arithmetic types and represent integer types or floating point values throughout the program. The predefined architecture-independent integer types are: `int`, `int8`, `int16`, `int32`, `int64` (signed integers) and `uint`, `uint8`, `uint16`, `uint32`, `uint64` (unsigned integers), `float32` and `float64`.
 - `int`: This represents 32- or 64-bit integers, depending on the underlying platform. Size and other details are the same as that of C language.
 - Other numeric types include `byte` and `rune`. The former is an alias of `uint8` and `rune` is an alias of `int32`.
3. String types represent the set of string values. Its value is a sequence of bytes. Strings are immutable types, the contents of which cannot be changed once created. The pre-declared string type is `string`.
4. Derived types, which include pointer types, array types, structure types, union types, function types, slice types, interface types, map types and channel types.

Variable declaration in Go

The `var` name type is the syntax to declare a single variable. For example: `var a int`, `var b bool`, `var st string`.

```
package main
import "fmt"
func main()
```